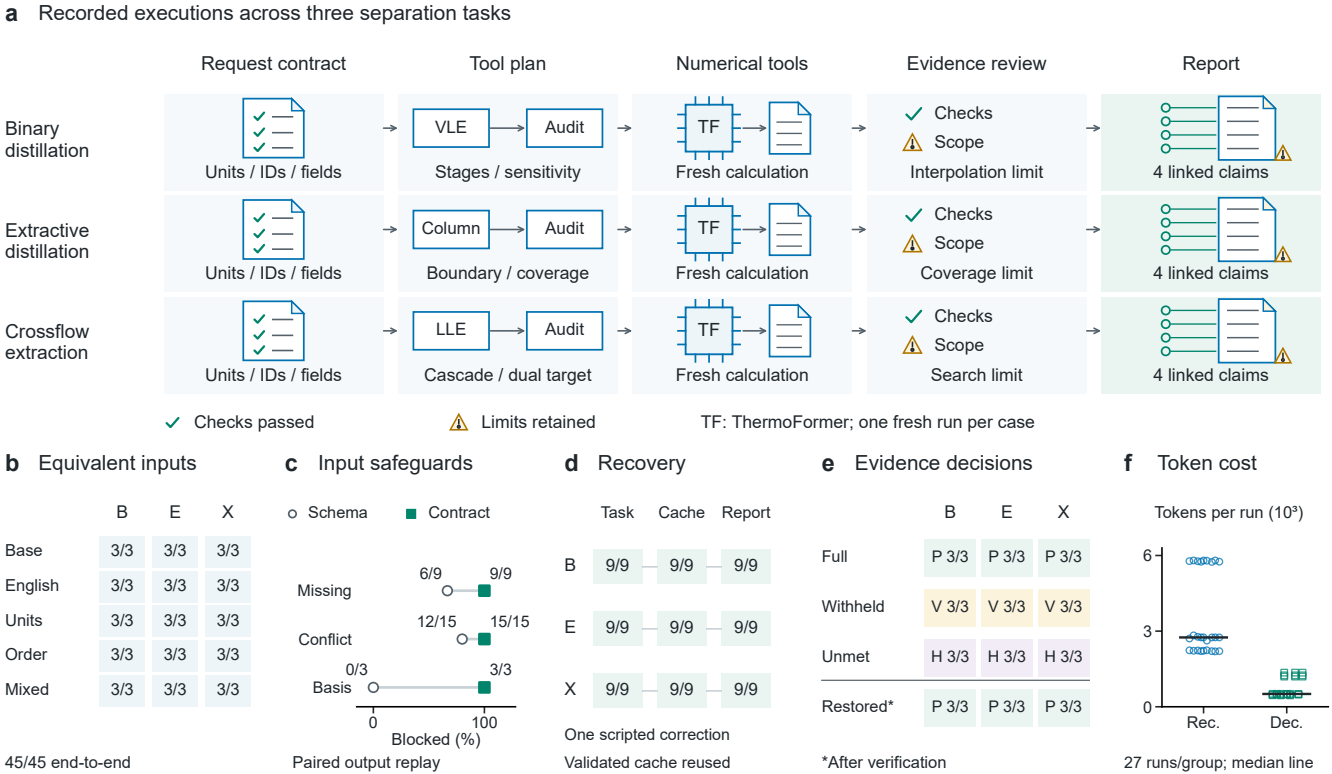

a Recorded executions across three separation tasks
Request contract
Tool plan
Numerical tools
Evidence review
Report
Binary distillation
Extractive distillation
Crossflow extraction
Units / IDs / fields
Units / IDs / fields
Units / IDs / fields
VLE
Audit
Column
Audit
LLE
Audit
Stages / sensitivity
Boundary / coverage
Cascade / dual target
TF
TF
TF
Fresh calculation
Fresh calculation
Fresh calculation
Checks
Scope
Interpolation limit
Checks
Scope
Coverage limit
Checks
Scope
Search limit
4 linked claims
4 linked claims
4 linked claims
Checks passed
Limits retained
TF: ThermoFormer; one fresh run per case
b Equivalent inputs
B E X
Base 3/3 3/3 3/3
English 3/3 3/3 3/3
Units 3/3 3/3 3/3
Order 3/3 3/3 3/3
Mixed 3/3 3/3 3/3
45/45 end-to-end
c Input safeguards
Schema
Contract
Missing
6/9
9/9
Conflict
12/15
15/15
Basis
0/3
3/3
0
100
Blocked (%)
Paired output replay
d Recovery
Task Cache Report
B 9/9 9/9 9/9
E 9/9 9/9 9/9
X 9/9 9/9 9/9
One scripted correction
Validated cache reused
e Evidence decisions
B E X
Full P 3/3 P 3/3 P 3/3
Withheld V 3/3 V 3/3 V 3/3
Unmet H 3/3 H 3/3 H 3/3
Restored* P 3/3 P 3/3 P 3/3
*After verification
f Token cost
Tokens per run (10³)
6
3
0
Rec.
Dec.
27 runs/group; median line
B: binary; E: extractive; X: crossflow
P: report with limits; V: verify; H: withhold target claim
Rec.: recovery; Dec.: decision